MIL-I-60507 (MU)
13 June 1966
SUPERSEDING
FA-PD-MI-2560, Rev 0
17 August 1962

MILITARY SPECIFICATION
INITIATOR, CARTRIDGE ACTUATED, DELAY, XM64, ASSEMBLY

1. SCOPE

1.1 This specification covers Initiator, Cartridge Actuated, Delay, XM64, Assembly loaded with Cartridge, Initiator, Delay, XM148, for use in conjunction with other propellant actuated devices in the recovery system of training bombs, or in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION
Military
MIL-A-2550 - Ammunition and Special Weapons, General Specification For

STANDARDS
Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1167 - Ammunition Data Card
MS-28741 - Hose Assembly, Detachable End Fitting, Medium Pressure

DRAWINGS
Munitions Command
F10522498 - Initiator, Cartridge Actuated, Delay, XM64 Assembly
C8846785 - Container, Ammunition Fiber M459 for Initiators, Delay, Cartridge Actuated
C8846786 - Box, Packing, for Initiators, Delay Cartridge Actuated in Fiber Container M459

FSC 1377

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2.1 (cont)

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications and the following:

3.2 Firing pin release.- The firing pin of the assembled initiator, less safety pin and cartridge, shall release from the cocked position when the application of an axial load of 15 to 30 pounds has withdrawn the initiator pin .370 inch, minimum, and .425 inch, maximum.

3.3 Initiator pin security.- Application of an axial load of 200 to 220 pounds to the initiator pin of the assembled initiator shall not withdraw the initiator pin past the safety pin. Load application shall cause no permanent deformation of the safety pin.

3.4 Breakaway torque.- The chamber and cap of the assembled initiator shall withstand a breakaway torque application of 400 inch pounds without loosening or deformation.

3.5 Residual magnetism.- The initiator shall not deflect the indicator of the compass more than three degrees in either direction when passed in front of the compass at a distance of six inches.

3.6 Functional.

3.6.1 Ballistics.- Initiators loaded with accepted cartridges and temperature conditioned shall produce pressure values specified in table I. Pressure values shall be produced at the end of three feet of aircraft hose, size No. 4, as specified in MS-28741.

Table I

Temperature	Pressure (psi) Min	Max	Ignition Delay (see 6.2)
-65° to -70° F	1300	2200	0.90 second, max
65° to 75° F	1500	2350	0.75 ± 0.15 second
160° to 165° F	1550	2450	0.60 second, min

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15. Page 12, paragraph 3.14: Add the following:

"Where specific tests to be performed during each service condition test are listed, satisfactory completion of such tests shall be considered as compliance with the service condition requirement."

16. Page 13, paragraph 3.14.2, replace with:

"3.14.2 Temperature (see 4.55).-

a. Operating: Ambient temperature in the range of 149°F to -40°F (The 149°F temperature includes effect of sun-load). Exposure at the high and low temperature extreme shall be 24 hrs min. At steps 3 and 8, the power shall be applied and receiver remain energized for 24 hrs min. Gain, power output and selectivity shall then be measured and recorded and shall meet the requirements of 3.13.6, 3.13.32 and 3.13.38.

b. Non-Operating: Exposure at the high (160°F and low -80°F) temperature extreme shall be 72± 4 hrs."

17. Page 13, paragraph 3.14.7, replace with:

"3.14.7 Vibration.- Except for resonance of specified parts and subassemblies, the equipment shall have no mechanical resonance below 55 cycles per second. (See 4.51)"

18. Page 13, paragraph 3.14.7.1: Delete.

19. Page 14, paragraph 3.16, replace with: "The contractor shall furnish two (2) preproduction samples of equipment on order for approval, if required by the invitation for bids and contract. (4.3)."

20. Page 15, paragraph 4.1: Replace with:

"4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements."

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21. Page 15, paragraph 4.3:

a. In line 7 delete "and (5) immersion".

b. Add the following to Table II, Preproduction Inspection:

"Turning Knob Torque Test" (under inspection), "3.12.2" (under Requirement Paragraph), "4.54" (under Test Paragraph).

c. Delete the following from Table II, Preproduction Inspection:

Operating and storage temperature	3.14	4.55
Moisture resistance test	3.14	4.57
Mobile frequency stability	3.13.25	4.50

22. Page 16, paragraph 4.4, under "item" and "where required," add:

"Filter, radio interference 3.12.3"

23. Page 16, paragraph 4.5.1: In Table III, under "Visual and Mechanical" delete the first seven lines and replace with:

"R. F. and Crystal-Osc. Sub-chassis	3.13	4.8	4.0 dphu	15.0 dphu
I. F. Sub-chassis			1.5 dphu	6.5 dphu
VFO Sub-chassis			1.0 dphu	4.0 dphu
A. F. Sub-chassis			2.5 dphu	10.0 dphu
Power Supply Sub-chassis			1.0 dphu	4.0 dphu
Completed Receiver R-390( )/URR			1.5 dphu	6.5 dphu"

24. Page 17, Table III, after overall selectivity add: "Room ambient only."

25. Page 18, Table IV, delete: "Electro-mechanical filters".

26. Page 18, Table IV - Group B inspection, in "Test Paragraph" column; add: "and 4.61.1" to "4.61."

27. Page 18, paragraph 4.5.3.1 Sampling procedures for Group C Inspection.- Replace with the following:

"One receiver for each sub-group in Table IV shall be selected from the first units of production, without regard to quality, by the Government Inspector. Thereafter, one receiver for sub-groups, 1, 2, and 3 shall be selected for each 200, or portion thereof, produced; one receiver for sub-group 4 from each 500 units, or portion thereof".

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4.2.2.3 Assembly, Dwg F10522498, less Dwgs D10522709, C8597928, B10520494, C10520414 and B8359536.

Defect	Method of Inspection
CRITICAL	
1. Firing pin not cocked	Manual/Visual
2. Any part missing	Visual
3. Improper assembly	Visual
MAJOR	
101. Safety pin loose or deformed	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.4 Assembly, Dwg F10522498, less all drawings, except C8597928, C10520414 and D10522709.

CRITICAL	
1. Cartridge not properly seated	Visual
2. Filter missing or insecure	Visual
3. Nylon patch missing or improperly applied	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.5 Assembly, Dwg F10522498.

CRITICAL	
1. Cartridge face not seated against base of firing pin housing	X-ray
2. Any part missing, except as otherwise classified	X-ray
3. Improper assembly	X-ray
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Set screw missing or insecure	Visual
102. Burr	Visual
103. Foreign matter	Visual
MINOR	
201. Thread sealing compound missing from set screw	Visual
202. Shipping plug missing	Visual
203. Protective finish inadequate or defective	Visual

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4.2.2.6 Fiber Container, Dwg C8846785.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Gap between body and cover, max	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual
205. Marking missing, incorrect or illegible	Visual
206. Tape incomplete or badly wrinkled	Visual
207. Length of tear tab, min	Visual
208. Color of tape incorrect	Visual

4.2.2.7 Packing Box, Dwg C8846786.

CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual/Visual
203. Strapping missing, loose, broken or improperly assembled	Manual/Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with 4.3.

4.2.3.1 Firing pin release.- The initiator pin of each subassembly, consisting of all parts, except the safety pin, cartridge and chamber, shall be subjected to the firing-pin release test. Failure of any subassembly to comply with the requirement of 3.2 shall be cause for rejection of the subassembly.

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4.2.3.2 Initiator pin security.- The initiator pin of each subassembly consisting of all parts, except the chamber and cartridge, shall be subjected to the initiator-pin security test. Failure of any subassembly to comply with the requirement of 3.3 shall be cause for rejection of the subassembly.

4.2.3.3 Breakaway torque.- The initiator chamber and cap for each assembled initiator shall be subjected to the breakaway torque application. Failure of any assembly to comply with the requirement of 3.4 shall be cause for rejection of the assembly.

4.2.3.4 Functional.- The following tests shall be performed by the Government:

4.2.3.4.1 Ballistics.- Twenty-one units shall be tested as shown in figure 2 to determine compliance with the requirements of 3.6.1, 3.6.2 and 3.6.3. The units shall be temperature conditioned as provided in table III and 4.3.5.1.1. The variables sampling plan of MIL-STD-414, example C2 or C3, as applicable, for the AQL listed in table III shall be used for each sample of seven units. Failure of any unit to comply with the requirements of 3.6.1, 3.6.2 and 3.6.3, or failure of a sample to meet the criteria of the variables sampling plan shall be cause for rejection of the lot.

Table III

Temperature	Sample Size	AQL
-65° to -70° F	7	0.65%
65° to 75° F	7	0.65%
160° to 165° F	7	0.65%

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment approved by the Government.

4.3 Test methods and procedures.

4.3.1 Firing pin release.- The subassembly shall be secured in the fixture. A pull shall be applied to the initiator pin in the longitudinal direction, in conjunction with the device which registers the amount of pull.

4.3.2 Initiator pin security.- The subassembly shall be secured in the fixture with the safety pin in place. A pull shall be applied to the initiator pin in the longitudinal direction in conjunction with a device which registers the amount of pull. The test shall be performed with the safety pin in both positions.

4.3.3 Breakaway torque.- A torque wrench set at the torque requirement specified in 3.4 shall be used in conjunction with the device to determine the adequacy of the joint. The torque shall be applied to the cap.

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4.3.4 Residual magnetism.- The initiator shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the initiator six inches from and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading as shown in figure 1. The initiator shall be slowly moved past the compass in a longitudinal direction for its entire length rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90-degree rotation of the initiator.

4.3.5 Functional.

4.3.5.1 Ballistics.- The test sample shall be rigidly supported in the test fixture. The sample shall be tested within five minutes after removal from the conditioning chamber (see figure 2). When this time limit is exceeded, the sample unit shall be reconditioned at the original temperature.

4.3.5.1.1 Temperature conditioning.- A standard XM64 Initiator loaded with a dummy XM148 Cartridge, containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be a minimum of 1-hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking - Level A.- Packing and marking shall be in accordance with dwgs C8846785 and C8846786.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

6. NOTES

6.1 A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Ignition delay is defined as the time that elapses between the firing of the pressure element and the beginning of the continuous pressure rise on the pressure time record.

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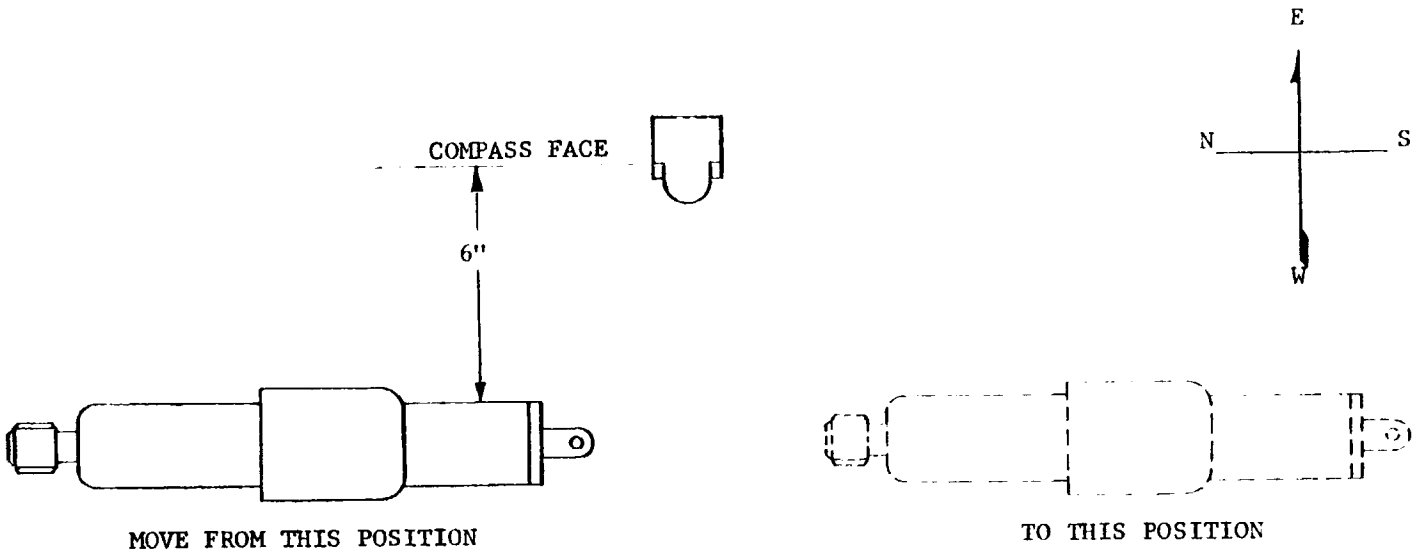
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PILOT'S STANDBY COMPASS, TYPE AN-5766



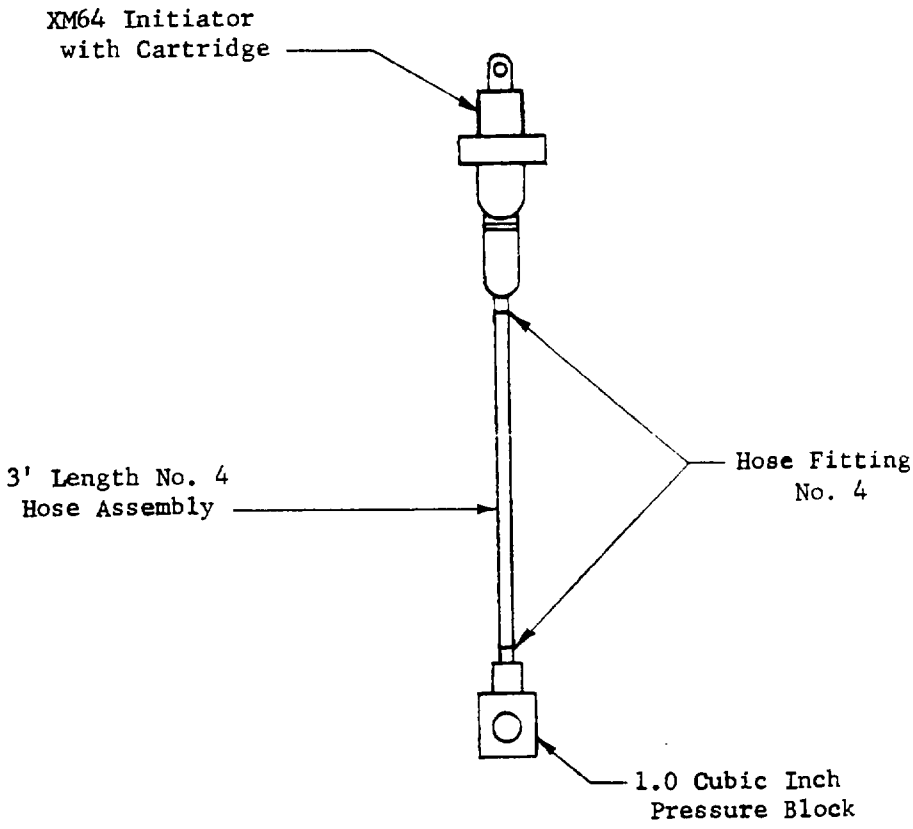
PLAN VIEW

NOTE: INITIATOR AND COMPASS SHALL LIE IN A HORIZONTAL PLANE

LAYOUT FOR DETERMINATION OF RESIDUAL MAGNETISM IN INITIATOR, CARTRIDGE ACTUATED, XM64

FIGURE 1

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TOP VIEW

PERFORMANCE TEST LAYOUT SKETCH

FIGURE 2

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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